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Algebraic Collapse in Nondeterministic Language Models via Deterministic Manifold Constraint

A Geometric-Calculus Framework for Hybrid Cognitive Architecture Safety

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Abstract

Large language models (LLMs) are, by construction, nondeterministic systems: at each decoding step, the next token is drawn from a probability distribution parameterized by the model's learned weights and a conditioning context. While this stochasticity endows LLMs with remarkable generative flexibility, it simultaneously renders their outputs formally unverifiable, resistant to formal methods, and unsafe for deployment in high-assurance cognitive systems without additional architectural intervention. This paper introduces and formalizes the notion of *algebraic collapse*: the phenomenon by which the effective output set of a stochastic language model, when filtered through a *deterministic manifold constraint layer*, is reduced from an infinite, continuous probability distribution to a finitely-generated algebraic structure — specifically, a structure inheriting the Lie

group properties of the constraint manifold.

We model the LLM output as a stochastic field $F: \Omega \times T \rightarrow \mathbb{R}^n$ and the constraint as a smooth projection $\Phi: \mathbb{R}^n \rightarrow \Sigma$ onto a compact Riemannian manifold $\mathcal{M} = (M, g)$ with deterministic constraint surface Σ . We prove the Algebraic Collapse Theorem: the constrained output set $\mathcal{A} = \Phi(\text{Image}(F))$ forms a finitely-generated algebraic structure whose generators are determined by the geodesic basis of Σ . We further derive Lyapunov stability conditions for the constrained dynamical system, establish fixed-point convergence via the Banach contraction principle, and bound the nondeterministic residual.

These results are instantiated in the *Hybrid Hemisphere Architecture*: a two-module cognitive system in which a deterministic manifold engine (Left Hemisphere) and a generative LLM (Right Hemisphere) are coupled through a Binding Layer implementing real-time algebraic collapse. Implications for AI safety, output verifiability, hallucination containment, and formal certification pathways are discussed. The framework positions alignment not as a behavioral engineering problem, but as a *manifold design problem*, offering a principled geometric foundation for safe generative cognition.

Keywords: *algebraic collapse, deterministic manifold constraint, nondeterministic language models, Riemannian geometry, Lie group, hybrid cognitive architecture, AI safety, Lyapunov stability, geodesic projection, neurosymbolic AI*

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1. Introduction

1.1 Motivation

The deployment of large language models in cognitive systems, autonomous agents, and decision-support infrastructure has accelerated dramatically over the first half of the 2020s. These systems achieve remarkable performance on tasks ranging from natural language understanding to mathematical reasoning, code generation, and multi-modal inference. Yet their fundamental computational character — stochastic, nondeterministic, and opaque — places

them in direct tension with the requirements of safety-critical engineering. In avionics, autonomous vehicles, medical decision support, and legal reasoning systems, outputs must be formally verifiable, reproducible, and traceable. A system that produces different outputs on identical inputs, and whose output space cannot be finitely characterized, cannot be certified under existing formal safety frameworks such as DO-178C for airborne software or ISO 26262 for automotive functional safety.

The root of this tension lies not in LLM capability but in LLM architecture. At each generation step, the model computes a probability distribution over a vocabulary of tens of thousands of tokens, samples from that distribution (introducing stochastic variation), and appends the result to the growing context. This process is inherently nondeterministic: even under zero temperature, floating-point non-associativity across hardware introduces variation; at any positive temperature, outputs are drawn from a proper distribution. The effective output space — the set of all strings the model might produce — is, in principle, unbounded and continuous. Formal verification methods, which require finite, enumerable state spaces, cannot be directly applied.

The question this paper addresses is therefore: *Can a nondeterministic LLM be architecturally coupled with a deterministic structure such that its effective output space collapses to a finitely-characterizable algebraic form, without sacrificing generative capacity?* We argue the answer is yes, and we provide the theoretical machinery to formalize and prove this claim.

1.2 The Core Claim: Algebraic Collapse

The central thesis of this work is as follows. When the output distribution of a stochastic language model is passed through a *deterministic manifold constraint* — a smooth projection operator mapping model outputs onto a compact Riemannian manifold carrying Lie group structure — the resulting constrained output set inherits the algebraic properties of the manifold. Specifically, the image of the projection is finitely generated, closed under a binary operation

induced by the group structure, and formally enumerable. We term this phenomenon **algebraic collapse**.

Algebraic collapse does not eliminate nondeterminism; it structures it. The stochastic field continues to explore diverse output regions; however, those outputs are geometrically projected onto a manifold whose algebraic structure is known, fixed, and auditable. From the perspective of a downstream verification system, the effective output space has collapsed from an uncountably large continuous distribution to a finitely-generated algebra — hence the name.

The manifold is not a filter in the information-theoretic sense: it does not simply truncate low-probability outputs. Rather, it is a *topological deformation* of the output space, continuously mapping all possible stochastic outputs to their nearest algebraically expressible counterpart within the constraint surface Σ . The collapse is smooth, differentiable, and entropy-reducing.

1.3 The Hybrid Hemisphere Metaphor

We introduce and formalize the *Hybrid Hemisphere Architecture* as a concrete instantiation of algebraic collapse in a cognitive system. This two-module design draws an architectural metaphor from the functional specialization of the human cerebral hemispheres (Gazzaniga, 2000; Corballis, 2014):

- **The Left Hemisphere (LH)** module is a deterministic manifold engine: symbolic, rule-based, algebraically closed, and formally verifiable. It models prior domain knowledge as a fixed topological structure — the constraint manifold $\mathcal{M}$.
- **The Right Hemisphere (RH)** module is the nondeterministic LLM: generative, probabilistic, and context-sensitive. It produces rich, diverse, and creative outputs, but its output distribution is unbounded.
- **The Binding Layer (BL)** is the interface between LH and RH. It implements the projection Φ , performing algebraic collapse in real time,

mediating between the stochastic field of the RH and the algebraic constraint surface of the LH.

This metaphor is strictly architectural: we make no neuroscientific claims about the correspondence of artificial and biological cognition. The metaphor is useful insofar as it captures the functional duality of generative and constrained reasoning in cognitive systems.

1.4 Contributions

The principal contributions of this paper are:

1. **The Algebraic Collapse Theorem** (Theorem 1): A formal proof that the constrained output set $\mathcal{A} = \Phi(\text{Image}(F))$ forms a finitely-generated algebraic structure under a binary operation induced by the Lie group structure of the constraint manifold.
2. **A Geometric-Calculus Stability Framework**: Lyapunov stability conditions for the constrained dynamical system, gradient flow characterization on the manifold, and fixed-point convergence guarantees via the Banach contraction principle.
3. **Formal Entropy and Divergence Analysis**: Quantitative characterization of the entropy reduction achieved by manifold constraint, including bounds on projection residual over time.
4. **The Hybrid Hemisphere Architecture**: A concrete, implementable cognitive system design instantiating algebraic collapse, with detailed specification of the Binding Layer and its computational properties.
5. **Safety and Certification Implications**: A structured analysis of how algebraic collapse enables formal verification, hallucination containment, auditability, and alignment via manifold design.

1.5 Paper Structure

The paper proceeds as follows. Section 2 reviews background in stochastic LLM generation, Riemannian manifold theory, algebraic structures in machine learning, and hybrid architectures. Sections 3 and 4 develop the formal machinery for the deterministic manifold and the stochastic LLM field, respectively. Section 5 presents and proves the Algebraic Collapse Theorem and its corollaries. Section 6 provides the calculus-based stability analysis. Section 7 describes the Hybrid Hemisphere Architecture in full detail. Section 8 analyzes implications for cognitive system safety. Section 9 discusses open questions and broader implications. Section 10 concludes.

2. Background and Related Work

2.1 Stochastic Token Generation in Large Language Models

Modern autoregressive language models, introduced in their large-scale form by Vaswani et al. (2017) and scaled to billions of parameters by Brown et al. (2020), generate text by iteratively sampling from a conditional probability distribution over a fixed vocabulary V . At generation step t , given a token sequence context $c = (x_1, x_2, \dots, x_{t-1})$, the model computes a logit vector $z \in \mathbb{R}^{|V|}$ via a deep transformer network, and converts this to a probability distribution via the softmax operation:

$$P(x_t = v \mid c) = \exp(z_v / \tau) / \sum_{v' \in V} \exp(z_{v'} / \tau)$$

where $\tau > 0$ is the temperature parameter. For $\tau = 1$ (standard softmax), $\tau \rightarrow 0$ (greedy/deterministic decoding), and $\tau > 1$ (increased stochasticity). Even at $\tau = 0$, hardware-level floating-point non-associativity means that deterministic decoding is only guaranteed under tightly controlled execution environments (Northcutt et al., 2021).

Several sampling strategies further modulate the distribution. *Top-k sampling* (Fan et al., 2018) restricts the sampling pool to the top- k probability tokens; *nucleus (top-p) sampling* (Holtzman et al., 2020) samples from the smallest subset of tokens whose cumulative probability exceeds a threshold p . These strategies trade diversity for coherence but do not eliminate nondeterminism: they redistribute probability mass rather than collapsing it to a point.

The cumulative effect is that the output of an LLM over a generation trajectory of length T is a *stochastic process* — a sequence of random variables ($X_1, X_2, \dots, X_T$) drawn from a joint distribution that is high-dimensional, context-dependent, and in general, analytically intractable. The output space — the set of all possible output strings — has cardinality $|V|^T$, which for $|V| = 50,000$ and $T = 1000$ is astronomically large. This is the fundamental source of unverifiability.

2.2 Geometric Manifold Theory in Machine Learning

The application of differential geometry to machine learning has a rich history, originating in information geometry (Amari, 1998) and expanding into geometric deep learning (Bronstein et al., 2021). A *smooth manifold* M is a topological space that locally resembles Euclidean space $\mathbb{R}^n$, equipped with a differential structure permitting the definition of smooth functions, tangent vectors, and differential forms. A *Riemannian manifold* (M, g) additionally carries a smooth metric tensor g , defining inner products on tangent spaces and thereby enabling the measurement of lengths, angles, and curvatures.

In the context of representation learning, the *manifold hypothesis* asserts that high-dimensional data (images, text embeddings, audio) is concentrated near a low-dimensional smooth manifold embedded in the ambient high-dimensional space (Bengio et al., 2013). This hypothesis motivates dimensionality reduction and generative modeling approaches that explicitly model or exploit manifold structure.

For our purposes, the key geometric objects are: (1) the *tangent bundle* TM , the disjoint union of tangent spaces at each point; (2) the *geodesic*, the generalization of a straight line to a curved manifold, defined as the curve minimizing arc length between two points; and (3) the *Riemann curvature tensor* R , which measures the deviation of parallel transport from Euclidean behavior. These objects will play central roles in our formal framework.

Constraint manifolds have been studied in the context of robotics (Murray et al., 1994), control theory (Bullo and Lewis, 2004), and, more recently, in neural network weight-space geometry (Kaul, 2020). Projections onto constraint manifolds arise naturally in optimization under manifold constraints, a setting thoroughly studied by Absil et al. (2008) in the context of Riemannian optimization.

2.3 Algebraic Structures in Machine Learning

The study of algebraic structure in machine learning outputs is an emerging area with connections to formal verification, program synthesis, and theoretical computer science. A *group* is a set G equipped with a binary operation $\cdot$ satisfying closure, associativity, identity, and invertibility. A *ring* adds a second operation (typically addition) satisfying distributivity. A *semigroup* weakens the group axioms to closure and associativity only. These structures are relevant to ML outputs when the output set can be shown to be closed under some naturally induced operation.

Lie groups — smooth manifolds that are simultaneously groups, with smooth group operations — are particularly important here. They arise naturally as symmetry groups of differential equations and geometric structures (Lee, 2013). When a constraint manifold carries Lie group structure, the group operations are geometrically meaningful: the composition of two valid outputs (elements of the manifold) is itself a valid output (the manifold is closed under the group operation).

Prior work has explored algebraic structure in recurrent network attractors (Gori et al., 2004), in the weight symmetries of neural networks (Wood and Shawe-Taylor, 1996), and in the algebraic structure of attention mechanisms (Tsai et al., 2019). However, to our knowledge, no prior work has formalized the algebraic collapse of LLM output distributions under smooth manifold constraints in the terms developed here.

2.4 Hybrid Architectures and Binding Mechanisms

The coupling of deterministic symbolic systems with probabilistic neural networks has been a longstanding goal of artificial intelligence research, broadly surveyed under the term *neurosymbolic AI* (Mao et al., 2019; Marcus, 2020; Garcez and Lamb, 2023). Representative approaches include: (1) neural theorem provers that couple differentiable neural reasoning with classical logic (Evans and Grefenstette, 2018); (2) constraint satisfaction networks that enforce logical constraints on neural network outputs via differentiable penalty functions (Diligenti et al., 2017); (3) program synthesis systems that use neural components to guide search over a symbolic program space (Devlin et al., 2017).

The notion of a *binding layer* — an interface between neural and symbolic components — has been discussed in the context of cognitive architectures (Anderson et al., 2004; Laird, 2012) and neural-symbolic integration (Besold et al., 2017). However, existing binding mechanisms are typically implemented as learned neural interfaces or hard-coded rule systems, without a formal geometric or algebraic characterization of the transformation they perform. The present work provides the first such characterization via the language of differential geometry and Lie group theory.

Closely related is the literature on *constrained decoding* for LLMs (Hokamp and Liu, 2017; Post and Vilar, 2018; Lu et al., 2021), which enforces lexical or structural constraints on generation. While constrained decoding provides a practical engineering tool, it operates at the token level without geometric or

algebraic formalization. Our framework generalizes constrained decoding to the geometric setting, providing it with a rigorous theoretical foundation.

3. The Deterministic Manifold: Formal Definition

3.1 Manifold Construction

We begin by constructing the mathematical object at the center of our framework: the deterministic manifold $\mathcal{M}$. Let $n \in \mathbb{N}$ be the dimension of the ambient embedding space (corresponding, in practice, to the dimensionality of the LLM's output representation layer). We define:

Definition 1 (Deterministic Manifold).

A *deterministic manifold* is a tuple $\mathcal{M} = (M, g)$ where:

- (i) M is a smooth, compact, connected, orientable d -dimensional manifold ($d < n$) embedded in $\mathbb{R}^n$ via a smooth embedding $\iota: M \hookrightarrow \mathbb{R}^n$;
- (ii) g is a Riemannian metric tensor on M , defining a smooth, positive-definite, symmetric bilinear form $g_p: T_pM \times T_pM \rightarrow \mathbb{R}$ on each tangent space T_pM ;
- (iii) M carries a Lie group structure $G = (M, \cdot)$ such that the group multiplication $\mu: M \times M \rightarrow M$ and inversion $\iota: M \rightarrow M$ are both smooth maps.

Compactness of M is essential: it ensures that the image of the projection operator (defined in §3.2) is bounded, and provides the compactness argument needed in the proof of Theorem 1 (§5.4). Orientability ensures that integration over the manifold is well-defined, enabling the entropy computations of §4.2. The Lie group structure G is the source of algebraic collapse: it is the structure that the constrained output set inherits.

Within the deterministic manifold, we designate a *constraint surface* as the specific subspace to which LLM outputs are projected:

Definition 2 (Constraint Surface).

The *deterministic constraint surface* is a closed submanifold $\Sigma \subset M$ satisfying:

- (i) Σ is a Lie subgroup of G ;
- (ii) Σ is the zero set of a smooth function $h: M \rightarrow \mathbb{R}^k$ with $k = n - d$ (i.e., $\Sigma = h^{-1}(0)$) and dh has full rank on Σ ;
- (iii) Σ is connected and orientable.

The choice of Σ as the zero set of a smooth constraint function h is standard in constrained optimization on manifolds (Absil et al., 2008) and enables explicit computational implementation of the projection via gradient-descent on h .

[Conceptual Figure: Three nested geometric objects. Outer space: ambient $\mathbb{R}^n$, represented as an open cloud of points. Middle layer: compact Riemannian manifold M , represented as a torus-like surface. Inner layer: constraint surface Σ , shown as a closed curve on the torus. Arrows indicate the projection Φ mapping ambient-space points toward Σ .]

Figure 1. Hierarchical structure of the deterministic manifold framework. The stochastic LLM field F produces outputs in $\mathbb{R}^n$. The Riemannian manifold $\mathcal{M} = (M, g)$ is embedded in $\mathbb{R}^n$. The constraint surface $\Sigma \subset M$ carries Lie group structure G . The projection Φ maps $F(\omega, t)$ to its nearest point in Σ .

3.2 Geometric Constraints and the Projection Operator

The core mechanism of algebraic collapse is the *constraint projection operator* Φ , which maps arbitrary points in the ambient space $\mathbb{R}^n$ onto the constraint surface Σ . We define this as the metric projection:

Definition 3 (Constraint Projection).

The *constraint projection operator* is the map $\Phi: \mathbb{R}^n \rightarrow \Sigma$ defined by:

$$\Phi(\mathbf{x}) = \operatorname{argmin}_{\sigma \in \Sigma} d(\mathbf{x}, \sigma)$$

where $d(\mathbf{x}, \sigma) = \|\mathbf{x} - \mathbf{u}(\sigma)\|_2$ is the Euclidean distance between $\mathbf{x} \in \mathbb{R}^n$ and the embedded image of $\sigma \in \Sigma$. The projection is well-defined and unique on the *medial complement* of Σ — the open subset of $\mathbb{R}^n$ where nearest-point projections are unique.

The operator Φ is a retraction in the sense of Absil et al. (2008): a smooth map satisfying $\Phi(\sigma) = \sigma$ for all $\sigma \in \Sigma$ (idempotency) and $\|d\Phi_\sigma\| = 1$ at points of Σ (unit-norm differential). These properties ensure that valid outputs (those already on Σ) are passed through unchanged, and that the projection does not distort the manifold structure near Σ .

The *tangent bundle* $T\Sigma$ of the constraint surface is the disjoint union of tangent spaces:

$$T\Sigma = \sqcup_{\sigma \in \Sigma} T_\sigma \Sigma = \{(\sigma, \mathbf{v}) : \sigma \in \Sigma, \mathbf{v} \in T_\sigma \Sigma\}$$

A valid output trajectory must have its velocity vector at each point lie within the tangent bundle: for a curve $\gamma: [0,1] \rightarrow \Sigma$, the condition $\dot{\gamma}(t) \in T_{\gamma(t)}\Sigma$ for all t ensures the trajectory remains on the constraint surface. The projection operator Φ continuously maps output sequences into such valid trajectories.

3.3 Geodesic Paths as Deterministic Output Trajectories

Within the constraint surface Σ , the natural notion of a "straight path" is the *geodesic* — the locally length-minimizing curve. For our framework, geodesics represent the deterministic output trajectories of the constrained system: paths that are as "direct" as possible within the manifold, without departing from Σ .

Definition 4 (Geodesic Output Trajectory).

A smooth curve $\gamma: [0, 1] \rightarrow \Sigma$ is a *geodesic output trajectory* if it satisfies the geodesic equation:

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0$$

where ∇ is the Levi-Civita connection on $(\Sigma, g|_{\Sigma})$ — the restriction of the Riemannian metric to Σ . In local coordinates x^i , the geodesic equation expands to:

$$\frac{d^2 x^i}{dt^2} + \Gamma^i_{jk} \left(\frac{dx^j}{dt} \right) \left(\frac{dx^k}{dt} \right) = 0$$

where Γ^i_{jk} are the Christoffel symbols of the Levi-Civita connection.

The geodesic equation $\nabla_{\{\dot{\gamma}\}} \dot{\gamma} = 0$ is the intrinsic condition for parallel transport of the velocity vector along the curve: the curve has no intrinsic acceleration within the manifold. Physically, geodesics are the paths a constrained system follows when it is subject to no forces other than those maintaining the constraint. For our architecture, this corresponds to the natural, "inertial" output trajectory of the deterministic manifold engine.

For a compact Lie group (such as the constraint surface Σ with its inherited group structure), the exponential map $\exp_{\sigma}: T_{\sigma}\Sigma \rightarrow \Sigma$ maps tangent vectors to geodesics:

$\gamma(t) = \exp_{\sigma}(tv)$ for initial point $\sigma \in \Sigma$ and initial velocity $v \in T_{\sigma}\Sigma$. The geodesics of a compact Lie group are the one-parameter subgroups — curves of the form $\gamma(t) = \sigma \cdot \exp(tX)$ for X in the Lie algebra $\mathfrak{g}$ of G . This will be critical in §5.4, where the finite set of such directions provides the generators of the algebraic structure.

3.4 Curvature and Stability

The *Riemann curvature tensor* R encodes the deviation of the manifold from flat Euclidean geometry. For vector fields X, Y, Z on Σ , the curvature tensor is defined by:

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z$$

where $[X, Y]$ is the Lie bracket of vector fields. The *sectional curvature* $K(X, Y)$ of the 2-plane spanned by X and Y is:

$$K(X, Y) = g(R(X, Y)Y, X) / (g(X, X)g(Y, Y) - g(X, Y)^2)$$

Curvature has direct implications for the stability of output trajectories on the manifold. In regions of **low sectional curvature** ($K \approx 0$), nearby geodesics diverge slowly; outputs in these regions are *stable* — small perturbations in the input produce small, predictable changes in the projected output. In regions of **high curvature** ($K \gg 0$), geodesics diverge rapidly (positive curvature) or converge to conjugate points (negative curvature), producing *instability* in the projected outputs. We call this effect *nondeterminism leakage*: the manifold constraint fails to fully suppress output variation in high-curvature regions.

Remark 1 (Curvature-Stability Correspondence).

For the constraint surface Σ endowed with the bi-invariant metric of a compact Lie group G , the sectional curvature satisfies $K(X, Y) = (1/4) \|[X, Y]\|^2 \geq 0$. Hence Σ has non-negative sectional curvature everywhere, and the flat regions ($K = 0$) correspond to commuting directions in the Lie algebra $\mathfrak{g}$ — precisely the directions

along which output variation is perfectly suppressed. This identifies the center of g as the maximally stable output subspace.

4. The Nondeterministic LLM as a Stochastic Field

4.1 Output Distribution Formalization

We model the output of the nondeterministic LLM as a *stochastic field* — a family of random variables indexed by a token sequence parameter — over the ambient embedding space $\mathbb{R}^n$.

Definition 5 (LLM Stochastic Field).

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space (where Ω is the sample space encoding all sources of randomness in the LLM: weight randomness, sampling randomness, hardware non-determinism) and let $T = \{1, 2, \dots, T_{\max}\}$ be the index set of generation steps. The *LLM stochastic field* is a measurable map:

$$\mathbf{F}: \Omega \times T \rightarrow \mathbb{R}^n$$

such that for each fixed $t \in T$, $F(\cdot, t): \Omega \rightarrow \mathbb{R}^n$ is a random variable representing the model's output embedding at step t . The output at each step follows the conditional distribution $X_t = F(\omega, t) \sim P(\cdot | c_{<t})$ where $c_{<t}$ is the context at step t .

The stochastic field formalization captures the full generality of LLM generation: different samples $\omega \in \Omega$ correspond to different decoding trajectories (i.e., different random seeds, different sampling outcomes at each step), and the map

$F(\omega, \cdot): T \rightarrow \mathbb{R}^n$ traces the embedding-space trajectory of a single generation run. The *image* of the stochastic field is:

$$\text{Image}(F) = \{F(\omega, t) : \omega \in \Omega, t \in T\} \subseteq \mathbb{R}^n$$

This is the set of all embedding-space points reachable by any generation trajectory of the LLM. In general, $\text{Image}(F)$ is a high-dimensional, non-convex, possibly disconnected subset of $\mathbb{R}^n$, without any intrinsic algebraic structure.

4.2 Entropy and Divergence Characterization

The *Shannon entropy* of the LLM output at step t , with respect to a reference measure μ on $\mathbb{R}^n$, is:

$$H(X_t) = -\int_{\mathbb{R}^n} p_t(x) \log p_t(x) d\mu(x)$$

where p_t is the density of X_t . High entropy corresponds to high uncertainty in the output — the LLM is exploring a large region of the output space. In practice, the distribution $P(\cdot|c_{<t})$ has entropy controlled by the softmax temperature τ : as $\tau \rightarrow 0$, entropy decreases; as $\tau \rightarrow \infty$, entropy increases to $\log|V|$.

The *KL divergence* between the unconstrained and constrained output distributions quantifies the information cost of applying the manifold constraint:

$$D_{\text{KL}}(P_{\text{constrained}} \parallel P_{\text{unconstrained}}) = \int_{\Sigma} p^{\Phi}_t(\sigma) \log[p^{\Phi}_t(\sigma) / p_t(\iota(\sigma))] d\text{Vol}_{\Sigma}(\sigma)$$

where p^{Φ}_t is the pushforward of p_t under Φ and $d\text{Vol}_{\Sigma}$ is the Riemannian volume element on Σ . The KL divergence is always non-negative and equals zero if and only if the LLM output distribution is already supported on Σ — i.e., when no projection is needed. This provides a natural measure of the "mismatch" between the LLM's natural output distribution and the constraint manifold.

4.3 The Projection Problem

The central computational challenge of the framework is the *projection problem*: given a realization $F(\omega, t) \in \mathbb{R}^n$ of the stochastic field, compute $\Phi(F(\omega, t)) \in \Sigma$ efficiently and accurately. In general, exact metric projection onto a nonlinear manifold is a nonlinear optimization problem:

$$\Phi(F(\omega, t)) = \operatorname{argmin}_{\{\sigma \in \Sigma\}} \|F(\omega, t) - \iota(\sigma)\|^2$$

We define the *projection residual* as the distance by which the raw LLM output departs from the constraint surface:

Definition 6 (Projection Residual).

The *projection residual* at step t for outcome $\omega \in \Omega$ is:

$$\varepsilon(\omega, t) = \|F(\omega, t) - \Phi(F(\omega, t))\|_2$$

The *expected projection residual* at step t is $E_\omega[\varepsilon(\omega, t)]$. The *residual process* $\{\varepsilon(\omega, t)\}_{t \in T}$ measures the ongoing departure of the stochastic field from the constraint surface over the generation trajectory.

The residual $\varepsilon(\omega, t)$ is zero when the LLM output already lies on Σ and positive otherwise. The Binding Layer's primary function is to drive ε to zero by applying Φ in real time. The behavior of the residual process over time — and its convergence to zero — is the subject of §6.

[Conceptual Figure: Two panels. Left panel: unconstrained LLM output trajectories, shown as stochastic paths wandering through a high-dimensional space — scattered, diverging, without clear structure. Right panel: the same trajectories

after projection Φ , each path snapping to the constraint surface Σ and following geodesic curves — ordered, bounded, algebraically structured.]

Figure 2. Effect of the constraint projection Φ on LLM output trajectories. Left: unconstrained stochastic field $F(\omega, t)$ with high projection residual ε . Right: projected field $\Phi(F(\omega, t))$ confined to the constraint surface Σ , following geodesic trajectories with $\varepsilon \rightarrow 0$.

5. Algebraic Collapse: The Central Theorem

5.1 Informal Statement

The central theoretical claim of this paper can be stated informally as follows. A nondeterministic language model, left to its own devices, produces outputs drawn from an enormously large and structurally opaque probability distribution. When these outputs are projected onto a deterministic manifold carrying algebraic (Lie group) structure, something remarkable happens: the projected output set inherits the algebraic structure of the manifold. The infinity of possible LLM outputs collapses, under projection, to a finite, structured, enumerable algebra. This is algebraic collapse.

To see why this is non-trivial, note that the *projection* of an arbitrary set onto a structured space does not in general produce a structured set: projection onto a Lie group would not, for instance, make a random point cloud algebraically closed. The key insight is that (a) the compactness of Σ bounds the projected set, (b) the smoothness of Φ preserves enough structure from the group operations, and (c) the Lie group structure of Σ provides the algebraic framework into which the projected set fits.

5.2 Formal Setup

We now set up the formal machinery for the theorem. Given the stochastic field $F: \Omega \times T \rightarrow \mathbb{R}^n$ (Definition 5) and the constraint projection $\Phi: \mathbb{R}^n \rightarrow \Sigma$ (Definition 3), define:

Definition 7 (Constrained Output Set).

The *constrained output set* is:

$$\mathcal{A} = \Phi(\text{Image}(F)) = \{\Phi(F(\omega, t)) : \omega \in \Omega, t \in T\} \subseteq \Sigma$$

This is the set of all constraint-projected outputs of the LLM stochastic field. It is the effective output set of the constrained system — the set of all outputs the Binding Layer can produce from any possible LLM generation trajectory.

We define a binary operation $\oplus$ on $\mathcal{A}$ induced by the Lie group structure G of Σ :

$$a \oplus b = \Phi(\mu(a, b)) \text{ for } a, b \in \mathcal{A}$$

where $\mu: \Sigma \times \Sigma \rightarrow \Sigma$ is the group multiplication of G . Since $\mu(a, b) \in \Sigma$ (by closure of the group) and Φ is idempotent on Σ , we have $a \oplus b = \mu(a, b)$: the operation $\oplus$ on $\mathcal{A}$ reduces to the group multiplication, restricted to $\mathcal{A}$.

5.3 The Algebraic Collapse Theorem

Theorem 1 — Algebraic Collapse

Let $\mathcal{M} = (M, g)$ be a compact Riemannian manifold with deterministic constraint surface $\Sigma \subset M$ carrying a Lie group structure $G = (\Sigma, \cdot)$, endowed with the bi-invariant metric inherited from g . Let $\Phi: \mathbb{R}^n \rightarrow \Sigma$ be the smooth metric projection (Definition 3). Let $F: \Omega \times T \rightarrow \mathbb{R}^n$ be a stochastic field representing LLM output

(Definition 5). Define the constrained output set $\mathcal{A} = \Phi(\text{Image}(F))$ (Definition 7) and the binary operation $a \oplus b = \mu(a, b)$ for $a, b \in \mathcal{A}$.

Then:

- (i) **[Algebraic Closure]** $\mathcal{A}$ forms a sub-semigroup of G under $\oplus$.
- (ii) **[Finite Generation]** $\mathcal{A}$ is finitely generated: there exists a finite set $\mathcal{G} = \{\gamma_1, \dots, \gamma_k\} \subset \mathcal{A}$ of generators, determined by the geodesic basis of Σ , such that every element of $\mathcal{A}$ is expressible as a finite product under $\oplus$ of elements of $\mathcal{G}$.
- (iii) **[Entropy Reduction]** $H(\Phi(F)) \leq H(F)$, with equality if and only if Φ is the identity on $\text{Image}(F)$ — i.e., if the LLM outputs already lie in Σ .
- (iv) **[Enumerability]** $\mathcal{A}$ is countably generated from the Lie algebra $\mathfrak{g}$ of G , and hence formally enumerable.

5.4 Proof Sketch

Proof Sketch of Theorem 1.

We proceed through each part.

(i) Algebraic Closure. By Definition 7, $\mathcal{A} \subseteq \Sigma$. Since Σ is a Lie subgroup of G , Σ is closed under the group multiplication μ . For any $a, b \in \mathcal{A} \subseteq \Sigma$, $a \oplus b = \mu(a, b) \in \Sigma$. We must show $\mu(a, b) \in \mathcal{A}$ — i.e., that $\mu(a, b)$ is in the image of $\Phi \circ F$. Since $\mathcal{A}$ is defined as the image of Φ applied to $\text{Image}(F)$, and Φ is idempotent on Σ , we have $\Phi(\mu(a, b)) = \mu(a, b)$. The closure of $\mathcal{A}$ under $\oplus$ in Σ follows from the group axioms of G ; $\mathcal{A}$ is at least a sub-semigroup of G .

(ii) Finite Generation. Since Σ is compact and carries the structure of a compact Lie group, the Lie algebra $\mathfrak{g} = T_{\mathbf{e}}\Sigma$ (tangent space at the identity) is finite-dimensional, say $\dim(\mathfrak{g}) = k$. Choose an orthonormal basis $\{X_1, \dots, X_k\}$ of $\mathfrak{g}$. The corresponding one-parameter subgroups $\gamma_i(t) = \exp(tX_i)$ for $i = 1, \dots, k$ are geodesics through the identity $\mathbf{e} \in \Sigma$. By the exponential map surjectivity theorem for compact Lie groups (Lee, 2013), every element of Σ is expressible as a finite

product of elements of the form $\exp(tX_i)$. Since $\mathcal{A} \subseteq \Sigma$, every element of $\mathcal{A}$ is generated by $\mathcal{G} = \{\exp(X_i) : i = 1, \dots, k\}$. The set $\mathcal{G}$ is finite (it has exactly $k = \dim(\mathfrak{g})$ elements), completing the finite generation argument.

(iii) Entropy Reduction. The projection Φ is a measurable map from $\mathbb{R}^n$ to Σ . By the data processing inequality for Shannon entropy (Cover and Thomas, 2006), for any measurable function f and random variable X : $H(f(X)) \leq H(X)$. Since $\Phi(F)$ is a deterministic function of F , $H(\Phi(F)) \leq H(F)$. Equality holds iff Φ is injective almost everywhere on the support of F , which occurs iff $\text{Image}(F) \subseteq \Sigma$ (i.e., Φ is the identity on the support).

(iv) Enumerability. The generators $\mathcal{G} = \{\exp(X_i)\}$ form a finite set. All elements of $\mathcal{A}$ are finite products of generators. The set of all finite products of a finite set of group elements is countable (it is a quotient of the free monoid on $\mathcal{G}$, which is countable). Hence $\mathcal{A}$ is countable and formally enumerable. $\square$

$\square$

5.5 Corollaries

Corollary 1 — Output Verifiability.

The constrained output set $\mathcal{A}$ is enumerable and finitely presented. Consequently, any output $a \in \mathcal{A}$ can be formally verified as a member of $\mathcal{A}$ by checking membership in the sub-semigroup generated by $\mathcal{G}$ — a decidable problem in finitely-generated group theory (Dehn, 1911).

Corollary 2 — Entropy Reduction Bound.

Let $d = \dim(\Sigma)$ and $n = \dim(\mathbb{R}^n)$. Under mild regularity conditions on the projection Φ (Lipschitz continuity with constant L), the entropy reduction

satisfies: $H(F) - H(\Phi(F)) \geq (n - d) \cdot \log(1/L)$. This quantifies the information-theoretic cost of the manifold constraint as a function of codimension and projection smoothness.

Corollary 3 — Effective Determinism Bound.

For a projection residual threshold $\delta > 0$, define the δ -effective determinism set $\mathcal{A}_\delta = \{\sigma \in \Sigma : \varepsilon(\omega, t) < \delta \text{ for all } \omega, t\}$. Then $\mathcal{A}_\delta$ is a compact subset of Σ and, for δ sufficiently small, the constrained system's output is indistinguishable (up to a precision of δ) from a deterministic function of the input context — i.e., it is effectively deterministic.

[Conceptual Figure: A tree diagram. Root node: "LLM Stochastic Field $F(\omega, t)$ ". First branching: many stochastic output trajectories. Middle layer: "Projection Φ applied by Binding Layer". Post-projection: trajectories converge to a finite set of nodes labeled with group elements $\{g_1, g_2, \dots, g_k\}$. These nodes are connected by arrows labeled " $\oplus$ " (group multiplication). Caption indicates this is the finitely-generated algebraic structure $\mathcal{A}$]

Figure 3. Visualization of algebraic collapse. The infinite stochastic field F collapses under projection Φ to a finitely-generated algebraic structure $\mathcal{A}$, whose elements are connected by the Lie group operation $\oplus$. The k generators (corresponding to the geodesic basis of Σ) are shown as primary nodes; all other elements are finite products thereof.

6. Calculus-Based Stability Analysis

6.1 The Lyapunov Stability Condition

We model the constrained system — the Binding Layer applying Φ to the LLM stochastic field — as a discrete-time dynamical system on Σ . Let $\sigma_t = \Phi(F(\omega, t)) \in \Sigma$ be the constrained output at step t . The system evolution is:

$$\sigma_{t+1} = \Phi(F(\omega, t+1)) = T(\sigma_t)$$

where $T: \Sigma \rightarrow \Sigma$ is the stochastic transition operator. We establish stability of this system via a Lyapunov function:

Definition 8 (Lyapunov Function for the Constrained System).

A *Lyapunov function* for the constrained dynamical system is a smooth map $V: \Sigma \rightarrow \mathbb{R}_{\geq 0}$ satisfying:

- (i) $V(\sigma^*) = 0$ for some equilibrium point $\sigma^* \in \Sigma$ (e.g., the group identity $e \in G$);
- (ii) $V(\sigma) > 0$ for all $\sigma \neq \sigma^*$;
- (iii) $V(T(\sigma)) \leq V(\sigma)$ for all $\sigma \in \Sigma$ (the discrete-time Lyapunov decrease condition).

A natural candidate for the Lyapunov function is the squared Riemannian distance to the equilibrium:

$$V(\sigma) = (1/2) d_{\Sigma}(\sigma, \sigma^*)^2$$

where d_{Σ} is the geodesic distance on $(\Sigma, g|_{\Sigma})$. In the continuous-time limit, the decrease condition $V(T(\sigma)) \leq V(\sigma)$ becomes:

$$dV/dt = \nabla V \cdot \dot{x} \leq 0$$

where $\dot{x}$ is the velocity of the system trajectory. This condition is satisfied when the system trajectory follows the negative gradient of V — precisely the gradient

flow condition of §6.2. Physically, the Lyapunov function measures the "distance" of the constrained output from the manifold equilibrium, and the Lyapunov condition asserts that this distance decreases monotonically over time. The manifold constraint thus acts as an energy minimizer, driving constrained outputs toward stable, predictable regions of Σ .

6.2 Gradient Flow on the Constraint Surface

In the continuous-time formulation, the evolution of the constrained output $\sigma_t \in \Sigma$ follows the negative gradient flow of a loss functional $L: \Sigma \rightarrow \mathbb{R}$:

$$dx/dt = -\nabla_{\Sigma} L(x)$$

where $\nabla_{\Sigma} L$ is the *Riemannian gradient* of L on Σ : the unique tangent vector field satisfying $g(\nabla_{\Sigma} L, v) = dL(v)$ for all tangent vectors $v \in T\Sigma$. The loss functional L encodes the alignment objective — typically, a combination of the projection residual ε and a prior over preferred output regions of Σ :

$$L(\sigma) = \lambda_1 \cdot \varepsilon(\sigma)^2 + \lambda_2 \cdot V(\sigma)$$

for regularization weights $\lambda_1, \lambda_2 > 0$. The gradient flow equation $dx/dt = -\nabla_{\Sigma} L(x)$ is a first-order ODE on Σ ; by the Picard-Lindelöf theorem, it has a unique local solution for smooth initial data. For compact Σ , the solution is global (exists for all $t \geq 0$).

The covariant gradient ∇_{Σ} accounts for the curvature of Σ : it ensures that the gradient flow stays on the manifold rather than "cutting through" the ambient space. In local coordinates x^i on Σ , the covariant gradient is:

$$(\nabla_{\Sigma} L)^i = g^{ij} \partial L / \partial x^j$$

where g^{ij} is the inverse of the metric tensor — the standard formula for raising indices in Riemannian geometry. In regions of low curvature, this reduces

approximately to the Euclidean gradient; in high-curvature regions, the metric correction $g^{\{ij\}}$ plays a significant role.

6.3 Fixed-Point Convergence via the Banach Theorem

Under suitable contractivity conditions on the composed operator $\Phi \circ F$, repeated application of the constraint system converges to a fixed point — a stable, algebraically expressible output:

Theorem 2 — Fixed-Point Convergence (Banach)

Let (Σ, d_Σ) be a complete metric space (which it is, as a compact Riemannian manifold). Suppose the composed operator $T = \Phi \circ \bar{F}: \Sigma \rightarrow \Sigma$ (where $\bar{F}$ is the expected LLM field operator) satisfies the contraction condition:

$$d_\Sigma(T(x), T(y)) \leq k \cdot d_\Sigma(x, y) \text{ for all } x, y \in \Sigma, k \in [0, 1)$$

Then T has a unique fixed point $\sigma^* \in \Sigma$, and the iteration $\sigma_{\{n+1\}} = T(\sigma_n)$ converges to σ^* for any initial $\sigma_0 \in \Sigma$, with convergence rate:

$$d_\Sigma(\sigma_n, \sigma^*) \leq k^n / (1 - k) \cdot d_\Sigma(\sigma_1, \sigma_0)$$

The contraction condition $\|T(x) - T(y)\| \leq k\|x - y\|$ for $k < 1$ is a quantitative statement about the sensitivity of the constrained system to initial conditions. It requires that the LLM-plus-projection pipeline is not too sensitive: a small change in the starting point leads to an even smaller change in the next step's output. This is related to, but distinct from, the temperature parameter of the LLM: low temperature encourages contraction, while high temperature discourages it.

The fixed point σ^* is the algebraically stable output of the constrained system — the output to which repeated application of the system converges. It is a member of $\mathcal{A}$ and hence is algebraically expressible from the generators $\mathcal{G}$. The

convergence rate k^n is exponential in the number of steps, providing rapid convergence to the stable algebraic output.

6.4 Stability Bounds and Residual Decay

Combining the Lyapunov condition with the Banach convergence guarantee, we derive bounds on the expected projection residual over time:

Proposition 1 — Asymptotic Residual Vanishing

Under the Lyapunov condition (Definition 8) and the Banach contraction condition (Theorem 2), the expected projection residual satisfies:

$$\mathbb{E}[\varepsilon(\omega, t)] \leq C \cdot k^t$$

for a constant $C > 0$ depending on the initial distribution of $F(\omega, 0)$ and the manifold geometry. In particular, $\mathbb{E}[\varepsilon(\omega, t)] \rightarrow 0$ as $t \rightarrow \infty$, at an exponential rate. This means the nondeterministic component of the LLM output asymptotically vanishes: the constrained system becomes effectively deterministic in the long run.

Proposition 1 has a powerful practical interpretation: in a deployed Binding Layer system, the longer the system operates, the closer its outputs cluster to the fixed-point algebraic expression σ^* . Early in a generation trajectory, significant nondeterminism may leak through; over time, the Lyapunov energy minimization drives the constrained output toward the stable algebraic attractor. This provides a natural "warm-up" period characterization for the constrained system.

[Conceptual Figure: A graph with time t on the x-axis and expected residual

$\mathbb{E}[\varepsilon(\omega, t)]$ on the y-axis. Three curves are shown for different contraction constants $k = 0.3$ (steep decay), $k = 0.6$ (moderate decay), and $k = 0.9$ (slow decay). All curves converge asymptotically to zero. A horizontal dashed line marks the δ -effective determinism threshold from Corollary 3.]

Figure 4. Theoretical decay of the expected projection residual $\mathbb{E}[\varepsilon(\omega, t)]$ under the Lyapunov stability condition, for three values of the Banach contraction constant k . The horizontal dashed line indicates the δ -effective determinism threshold (Corollary 3). All trajectories satisfy $\mathbb{E}[\varepsilon(\omega, t)] \leq C \cdot k^t \rightarrow 0$.

7. The Hybrid Hemisphere Architecture

7.1 Architectural Overview

We now describe the *Hybrid Hemisphere Architecture* (HHA) — the concrete cognitive system design in which algebraic collapse is instantiated. The HHA consists of three primary components:

Component	Role	Formal Counterpart	Key Properties
Left Hemisphere (LH)	Deterministic manifold engine; symbolic, rule-based, formally verifiable reasoning	Constraint surface Σ with Lie group G	Algebraically closed; enumerable output space; formally verifiable; models prior domain knowledge
Right Hemisphere (RH)	Nondeterministic LLM; generative, probabilistic, context-sensitive reasoning	Stochastic field F : $\Omega \times T \rightarrow \mathbb{R}^n$	Generatively rich; high entropy; context-sensitive; not directly verifiable

Component	Role	Formal Counterpart	Key Properties
Binding Layer (BL)	Interface; applies algebraic collapse; mediates LH-RH interaction	Projection operator $\Phi: \mathbb{R}^n \rightarrow \Sigma$	Real-time projection; entropy gating; algebraic normalization; feedback loop controller

The LH is the system's *constraint authority*: it maintains the deterministic manifold $\mathcal{M}$, enforces domain knowledge (represented as the geometry of Σ), and produces outputs that are, by construction, algebraically expressible and formally verifiable. The RH is the system's *generative engine*: it produces diverse, context-sensitive, and semantically rich outputs that the LH alone could not generate. The BL is the critical interface that makes the two hemispheres compatible: it accepts the raw, nondeterministic output of the RH and transforms it, via the projection Φ , into an algebraically structured output that the LH can accept, verify, and act upon.

7.2 Data Flow and Processing Pipeline

The end-to-end processing pipeline of the HHA proceeds in six steps:

HHA Processing Pipeline

Step 1 — Input Reception (RH): The external input (query, prompt, sensor signal) is received by the Right Hemisphere. The RH prepares a context $c_{\{<t\}}$ incorporating prior conversation history, retrieved context, and system instructions.

Step 2 — Stochastic Generation (RH): The RH (a large language model with parameters θ) generates the stochastic output $F(\omega, t) \sim P(\cdot | c_{\{<t\}}; \theta)$. This output is a point in the high-dimensional embedding space $\mathbb{R}^n$ with high entropy — the raw, nondeterministic LLM output.

Step 3 — Algebraic Collapse (BL): The Binding Layer receives $F(\omega, t)$ and applies the projection Φ , computing $\sigma_t = \Phi(F(\omega, t)) \in \Sigma$. The BL simultaneously checks the entropy gate (Step 3a) and performs algebraic normalization (Step 3b).

Step 4 — Constraint Reception (LH): The Left Hemisphere receives the projected output $\sigma_t \in \mathcal{A} \subseteq \Sigma$. Because $\mathcal{A}$ is finitely generated and algebraically structured, the LH can verify $\sigma_t \in \mathcal{A}$ decidable and proceed with formal reasoning.

Step 5 — Verified Response Generation (LH): The LH applies its symbolic reasoning machinery to σ_t , producing a verified, algebraically-expressible response. This response is formally certified as lying within the algebraic structure $\mathcal{A}$.

Step 6 — Context Feedback (BL $\rightarrow$ RH): The LH output and verification certificate are fed back to the RH via the BL as additional context $c_{\{t+1\}} = c_{\{t\}} \cup \{\sigma_t, \text{cert}(\sigma_t)\}$. This feedback loop refines the RH's next-step distribution toward outputs that will require less projection — implementing the Lyapunov energy decrease at the architectural level.

7.3 The Binding Layer in Detail

The Binding Layer implements three distinct sub-operations in sequence:

(a) Topological Projection. The BL first applies the metric projection $\Phi(x) = \operatorname{argmin}_{\{\sigma \in \Sigma\}} d(x, \sigma)$. In practice, this is approximated by iterative gradient descent on the squared distance functional: starting from an initial guess $\sigma^{\{0\}}$ (e.g., the nearest stored group element), the BL iterates $\sigma^{\{k+1\}} = \sigma^{\{k\}} - \alpha \cdot \nabla_{\Sigma} d(F, \sigma^{\{k\}})^2$ until convergence. For well-trained manifolds, this converges in a small number of steps.

(b) Entropy Gate. The BL monitors the conditional entropy of the RH output distribution: $H_t = H(X_t | c_{<t})$. If H_t exceeds a threshold $H_{\{max\}}$, the BL rejects the RH output and requests resampling — instructing the RH to regenerate with a lower temperature or a more constrained sampling strategy. This entropy gate prevents high-uncertainty outputs from reaching the LH, even after projection, as high-entropy outputs correspond to projection residuals that may be too large for reliable algebraic normalization.

(c) Algebraic Normalization. After projection and entropy gating, the BL applies algebraic normalization: it maps the projected point $\sigma_t \in \Sigma$ to the nearest element of the discrete algebraic structure $\mathcal{A}$. Formally, this is the operation $\sigma_t \mapsto \operatorname{argmin}_{\{a \in \mathcal{A}\}} d_{\Sigma}(\sigma_t, a)$. Since $\mathcal{A}$ is finitely generated, this nearest-element search is over a countable set and can be efficiently implemented via breadth-first search over the Cayley graph of $(\mathcal{A}, \oplus)$ with generating set $\mathcal{G}$.

7.4 Neurological Analogy

The Hybrid Hemisphere Architecture draws its metaphorical inspiration from the well-documented functional lateralization of the human cerebral cortex. The left cerebral hemisphere is classically associated with analytical, sequential, language-structured, and rule-governed cognition; the right hemisphere with holistic, associative, creative, and context-sensitive cognition (Gazzaniga, 2000; Corballis, 2014). The corpus callosum — the dense white-matter tract connecting the two hemispheres — serves as a high-bandwidth binding interface, mediating the integration of the two modes of processing (Geschwind and Levitt, 2007).

Our architectural metaphor maps precisely: the LH module $\leftrightarrow$ analytical/rule-based left hemisphere; the RH module $\leftrightarrow$ creative/associative right hemisphere; the Binding Layer $\leftrightarrow$ corpus callosum. We emphasize that this is an *architectural metaphor only*: we make no claim that the HHA replicates, models, or approximates the mechanisms of biological hemispheric processing. The metaphor is useful precisely because it provides an intuitive vocabulary for

discussing the integration of deterministic and nondeterministic reasoning modes in a cognitive architecture.

What the metaphor does suggest, and what our theory supports, is a broader principle: *safe generative cognition requires the coupling of a constrained, algebraically structured reasoning component with a generative, probabilistic component*. Whether this coupling is implemented via the corpus callosum in biology or via the Binding Layer in our architecture is secondary to the architectural principle itself.

7.5 Scalability and Computational Complexity

The computational complexity of the Binding Layer is dominated by the metric projection step. For a manifold Σ of dimension d embedded in $\mathbb{R}^n$, exact projection requires solving a d -dimensional nonlinear optimization problem. With iterative gradient descent and convergence tolerance ϵ_{proj} , this requires $O(d / \epsilon_{\text{proj}})$ iterations per projection, each costing $O(n)$ time for the gradient computation — a total cost of $O(nd / \epsilon_{\text{proj}})$ per projection call.

Operation	Complexity	Notes
Topological Projection (Φ)	$O(nd / \epsilon_{\text{proj}})$	Iterative gradient descent on Σ ; can be parallelized over batch dimension
Entropy Gate	$O(V)$	Single pass over softmax distribution; V = vocabulary size
Algebraic Normalization	$O(\mathcal{G} \cdot r)$	BFS over Cayley graph to depth r = average word length in $(\mathcal{A}, \oplus, \mathcal{G})$
Lyapunov Monitoring	$O(d^2)$	Geodesic distance computation; dominated by metric tensor evaluation
Feedback Loop Update	$O(T_{\text{ctx}})$	Linear in context length T_{ctx} ; standard transformer context extension

For practical deployment, the projection step can be amortized via learned approximations: a shallow neural network trained to predict $\Phi(\mathbf{x})$ directly from $\mathbf{x}$, with the exact iterative projection used only for verification. This reduces the online complexity to $O(d^2)$ for the approximate projection, with periodic exact-projection audits. The latency trade-off is thus between projection accuracy (exact vs. approximate) and inference speed.

8. Implications for Cognitive System Safety

8.1 Verifiability

The most immediate safety implication of algebraic collapse is the restoration of *formal verifiability* to generative AI outputs. In standard LLM deployment, the output space is effectively infinite and continuous: formal methods (model checking, theorem proving, abstract interpretation) cannot be applied because they require finite, enumerable state spaces. The algebraic collapse theorem directly resolves this: by projecting LLM outputs onto the finitely-generated algebraic structure $\mathcal{A}$, the constrained output space becomes amenable to formal methods.

Concretely, given a safety property expressed as a predicate $P: \mathcal{A} \rightarrow \{\text{true}, \text{false}\}$, verifying that all constrained outputs satisfy P reduces to checking $P(a)$ for all $a \in \mathcal{A}$ — a finite computation. This is in direct contrast to the unconstrained setting, where verifying a property of the LLM output requires reasoning about the full distribution $P(\cdot|\mathbf{c})$, which is high-dimensional, continuous, and analytically intractable.

The verifiability guarantee is conditional on the correctness of the manifold construction: if $\mathcal{M}$ accurately encodes the domain's safety requirements, then $\mathcal{A}$

is the set of all safe outputs, and the HHA is verifiably safe. Manifold design is thus the critical engineering challenge — a point we return to in §9.

8.2 Geometric Containment of Hallucination

Hallucination in LLMs — the generation of outputs that are factually incorrect, internally inconsistent, or contextually inappropriate — is a manifestation of the stochastic field exploring regions of the output space that are distant from the factual and contextually valid region. In the HHA framework, hallucination is geometrically defined as: outputs $F(\omega, t)$ whose projection residual $\varepsilon(\omega, t) = \|F(\omega, t) - \Phi(F(\omega, t))\|$ is large — i.e., points far from the constraint surface Σ .

The Binding Layer's projection Φ provides a *geometric containment* of hallucination: all outputs, regardless of how far the RH stochastic field has wandered, are projected back onto Σ . The worst-case hallucination under the HHA is bounded by the maximum projection residual $\varepsilon_{\{\max\}} = \sup_{\{\omega, t\}} \varepsilon(\omega, t)$, which is finite by the compactness of $\mathcal{M}$ (all bounded subsets of $\mathbb{R}^n$ have finite diameter, and the LLM output distribution has compact support under any finite-precision computation).

Moreover, by Proposition 1, the expected hallucination (as measured by the expected projection residual) decays exponentially over the generation trajectory. The HHA thus provides both a worst-case bound and an asymptotic guarantee on hallucination.

8.3 Auditability and Interpretability

The finitely-generated structure of $\mathcal{A}$ provides natural audit trails for constrained LLM outputs. Each output $a \in \mathcal{A}$ can be expressed as a finite word in the generators $\mathcal{G} = \{\gamma_1, \dots, \gamma_k\}$: a sequence of group operations applied to the generators. This *generator expression* — the word representing a in the free group on $\mathcal{G}$ — provides a compact, human-readable, algebraically traceable description of how the output was constructed.

For interpretability, the generators $\mathcal{G}$ can be given semantic labels: if the constraint manifold Σ encodes a domain ontology (e.g., medical knowledge, legal reasoning, mathematical truth), each generator corresponds to a semantically meaningful "move" in the domain — an elementary inference step, a domain rule application, or an ontological relation traversal. The generator expression of an output then constitutes an interpretable trace of the reasoning process, making the system's behavior auditable at the algebraic level.

8.4 Safety Certification Pathways

Existing safety certification frameworks for safety-critical software — DO-178C for airborne systems, ISO 26262 for automotive systems, IEC 62443 for industrial control systems — require that system behavior be formally specified, verified against the specification, and traceable through the development lifecycle. These requirements have, until now, been impossible to satisfy for generative AI systems due to the infinite, unstructured nature of their output spaces.

The HHA framework opens a pathway to certification by providing: (a) a formal specification of the constrained output space as the algebraic structure $\mathcal{A}$; (b) a formal proof (Theorem 1) that all constrained outputs lie in $\mathcal{A}$; (c) a formal stability analysis (§6) characterizing the convergence of the system; and (d) a formal verifiability guarantee (Corollary 1) enabling model checking over $\mathcal{A}$. Together, these constitute the formal artifacts needed to support a certification case for a generative AI component — albeit one that would require new certification standards adapted to the geometric-algebraic framework.

8.5 Failure Modes and Limitations

Despite its theoretical strength, the HHA framework has several important failure modes and limitations that must be acknowledged:

- **Manifold Miscalibration.** The safety guarantees of the HHA depend entirely on the constraint manifold $\mathcal{M}$ accurately encoding the domain's

safety requirements. If the manifold is miscalibrated — encoding incorrect knowledge, excluding valid outputs, or including invalid ones — the algebraic collapse theorem provides guarantees about the wrong structure. Manifold design is a domain-expert task requiring careful validation.

- **High-Curvature Instabilities.** As noted in §3.4, high-curvature regions of Σ produce instability in the constrained output: small perturbations in the RH output lead to large changes in the projected output. In practice, these regions should be identified and either avoided (by adjusting the manifold geometry) or flagged for human review.
- **Binding Layer Latency.** The topological projection step (§7.3a) introduces inference latency proportional to $O(nd / \epsilon_{\text{proj}})$. For large-dimensional manifolds and tight projection tolerances, this may be prohibitive for real-time applications. Learned projection approximations (§7.5) can mitigate but not eliminate this cost.
- **Adversarial Projection Gaps.** An adversary with knowledge of the manifold geometry could craft inputs that exploit the projection gap — inputs designed to produce projected outputs in a specific, adversarially chosen region of $\mathcal{A}$. This is analogous to adversarial examples in standard neural networks and requires robustness analysis specific to the manifold geometry.
- **Expressiveness Trade-Off.** Algebraic collapse, by definition, reduces the output space. There may be valid, safe, and useful outputs of the LLM that lie outside $\mathcal{A}$. The framework trades generative expressiveness for verifiable safety — a trade-off that must be calibrated for each application domain.

9. Discussion

9.1 Revisiting the Central Claim

We have established that nondeterminism in large language models is not intrinsically incompatible with formal verifiability — provided that the system architecture includes a deterministic manifold constraint layer. The Algebraic Collapse Theorem (Theorem 1) formalizes this claim: the stochastic field of the LLM, projected onto a compact Riemannian manifold with Lie group structure, produces a finitely-generated algebraic structure. The Lyapunov stability analysis (§6) shows that this structure is approached exponentially in time. The Hybrid Hemisphere Architecture (§7) demonstrates how this theory can be instantiated in a deployable cognitive system.

What makes this result non-trivial is the combination of three properties: (a) the compactness of the constraint surface, which bounds the projected output set; (b) the smoothness of the projection operator, which preserves continuity and group operations; and (c) the Lie group structure of the constraint surface, which provides the algebraic framework into which the projected outputs fit. Remove any one of these properties — for instance, use a non-compact constraint surface, or a non-smooth projection, or a surface without group structure — and the algebraic collapse result fails.

9.2 Open Questions

Several important open questions remain:

Manifold Construction for Given Domains. The most pressing practical question is: how does one construct the constraint manifold $\mathcal{M}$ for a given application domain? The manifold must encode the domain's safety requirements geometrically — a non-trivial task requiring both domain expertise and differential geometric knowledge. One promising direction is to learn the manifold from certified domain examples, using techniques from manifold

learning (Tenenbaum et al., 2000; Roweis and Saul, 2000) adapted to the constrained setting.

Optimal Lie Group Structure. For a given domain, what Lie group structure on Σ provides the best trade-off between expressiveness (large $\mathcal{A}$, fine-grained output resolution) and verifiability (small $|\mathcal{G}|$, efficient enumeration)? This is a question in applied Lie theory with no known general answer. Domain-specific analyses are needed.

Online Manifold Learning. In dynamic domains, the appropriate constraint manifold may change over time as new knowledge is acquired or domain requirements evolve. Online algorithms for updating $\mathcal{M}$ without retraining the RH LLM or recomputing the full projection would be valuable. This connects to the literature on online Riemannian optimization (Bonnabel, 2013).

Empirical Validation. The present work is theoretical. Empirical validation — demonstrating algebraic collapse in a practical LLM-plus-constraint-layer system, measuring projection residuals, and quantifying hallucination containment — is an essential next step. We propose a benchmark suite based on formal domain ontologies (e.g., medical coding systems, legal taxonomies, mathematical proof structures) as natural candidates for constraint manifold instantiation.

9.3 Broader Implications for AI Alignment

The algebraic collapse framework has profound implications for the AI alignment problem. The dominant paradigm in AI alignment treats alignment as a *behavioral engineering problem*: specify desired behaviors, collect demonstrations or feedback, and use reinforcement learning from human feedback (RLHF) or related methods to shape the model's behavior toward the desired distribution (Christiano et al., 2017; Bai et al., 2022). This approach is effective but fragile: it depends on the quality of the feedback signal, can be

gamed by the model, and provides no formal guarantees about out-of-distribution behavior.

The algebraic collapse framework suggests a fundamentally different paradigm: *alignment as manifold design*. Rather than shaping behavior via feedback, encode the desired behavior geometrically — as the structure of the constraint manifold $\mathcal{M}$. The manifold is not learned from the model's behavior; it is designed by domain experts to capture the structure of valid, safe, and aligned outputs. The model is then constrained to operate within this geometric structure, with formal guarantees about its effective output space.

This paradigm has several attractive properties relative to behavioral alignment: (a) it provides formal guarantees, not just empirical evidence; (b) it is robust to adversarial inputs (the manifold constraint is applied at the output level, not the input level); (c) it supports incremental refinement (the manifold can be extended or refined without retraining the LLM); and (d) it separates the concerns of generative capability (the RH LLM, which can be any state-of-the-art model) and safety (the LH manifold, which is designed and certified independently).

9.4 A Philosophical Note: Structuring Nondeterminism

At a philosophical level, the algebraic collapse framework embodies a distinctive position on the relationship between nondeterminism and structure. The naive view is that nondeterminism and formal structure are incompatible: a system is either stochastic (and therefore structurally opaque) or deterministic (and therefore formally tractable). The algebraic collapse framework challenges this dichotomy.

Nondeterminism, in our framework, does not disappear under manifold constraint — the RH LLM continues to sample from its full distribution. What changes is the *structure of the nondeterminism's effects*: the stochastic variation of the RH is absorbed by the projection, and what emerges on the LH side is

structured, algebraic, and formally characterizable. Nondeterminism is not eliminated; it is *structured*. The manifold constraint provides the structuring principle.

This is analogous, mathematically, to the way a quotient map collapses a complex space to a simpler one: the complexity does not vanish, but the equivalence classes — the elements of the quotient — have a clean algebraic structure. The projection Φ plays the role of the quotient map; the constrained output set $\mathcal{A}$ plays the role of the quotient space. The infinite complexity of the LLM distribution is not destroyed but organized, into a form that formal reasoning can handle.

10. Conclusion

This paper has introduced and developed the theory of *algebraic collapse* in nondeterministic language models constrained by a deterministic manifold layer. Our central result — the Algebraic Collapse Theorem (Theorem 1) — establishes that the stochastic output field of a large language model, when projected onto a compact Riemannian manifold with Lie group structure, produces a finitely-generated algebraic structure. This structure is enumerable, formally verifiable, and inherits the group properties of the constraint manifold.

We have provided a comprehensive theoretical framework supporting this result: a formal model of the LLM as a stochastic field (§4), a rigorous construction of the deterministic manifold and constraint projection (§3), Lyapunov stability analysis and Banach fixed-point convergence guarantees for the constrained dynamical system (§6), and a concrete architectural instantiation in the Hybrid Hemisphere Architecture (§7). The safety implications — verifiability, hallucination containment, auditability, and certification pathways — have been analyzed in §8.

Restated informally: a nondeterministic language model, when coupled with a deterministic manifold constraint, stops being an opaque stochastic process and becomes, at the output level, an algebraic machine — a machine whose outputs lie in a finite, structured, formally characterizable algebra. This does not reduce the model's generative power; it channels that power through a geometric structure that formal methods can reason about. The nondeterminism is not eliminated; it is structured.

We call on the research community to pursue three directions of follow-on work. First, **empirical validation**: implement the Hybrid Hemisphere Architecture with a practical LLM and a domain-specific constraint manifold, measure projection residuals, verify algebraic closure, and quantify hallucination containment against baselines. Second, **open-source tooling**: release reference implementations of the Binding Layer — including the topological projection, entropy gate, and algebraic normalization components — as open-source libraries to facilitate community adoption and extension. Third, **standards engagement**: engage with safety certification bodies (DO-178C, ISO 26262, and emerging AI-specific frameworks) to develop certification pathways for HHA-compliant generative AI systems, using the algebraic output structure as the formal foundation for safety cases.

The Hybrid Hemisphere Architecture represents more than a single technical contribution: it is a template for a new class of cognitive AI systems in which the creative richness of generative models and the formal rigor of constrained symbolic systems are not in opposition, but in structured, algebraically mediated collaboration. In this sense, the framework points toward a future in which safe, verifiable, and generatively powerful AI is not a contradiction in terms, but a designable geometric property of the system's architecture.

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Algebraic Collapse in Nondeterministic Language Models via Deterministic Manifold Constraint

Leon | Independent Research | September 2026 | Preprint

Amendment to "Algebraic Collapse in Nondeterministic Language Models via Deterministic Manifold Constraint"

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Edition: v1.1 — Amendment Edition, September 30, 2026. Supersedes the v1.0 draft of this manuscript; v1.0 is preserved unaltered in version history per the append-only doctrine.

Method of this amendment: An author-directed mathematical audit of v1.0 (September 30, 2026) identified six defects in and around Theorem 1 and its corollaries. This amendment corrects the statement and proof of the central theorem, retracts two corollary claims, and relabels the applied claims of §8.2. All other content of v1.0 (Sections 2–4, 6, 7, and the background apparatus) is unaffected and stands as written.

Provenance: v1.0 was drafted with AI assistance; the audit that produced these corrections was author-directed and is disclosed here in full. This manuscript was prepared for publication under the open-book method: corrections via disclosed versioning, never silent edits.

A. The Corrected Central Theorem

The v1.0 theorem asserted that the constrained output set $\mathcal{A} = \Phi(\text{Image}(F))$ is a finitely-generated algebraic structure. This is not correct as stated: the raw image of the projection is not closed under the induced operation, and no finite generating set for it exists in general. The corrected statement below redefines the object of study as the algebra *generated by* the constrained outputs, and locates the true content of the collapse phenomenon in two properties that survive rigorous proof: **countability of the reachable algebra** and **containment of that algebra inside a designed lower-dimensional subgroup** (the caging property).

Hypotheses

(H1) Countable generation capacity. The realizable output set of the language model is at most countable. This holds for any discrete-token autoregressive model: the vocabulary is finite, outputs are finite-length strings over it, and the set of such strings is countable. Hence $\text{Image}(F)$ is at most countable, and so is the projected set $\mathcal{A}_0 = \Phi(\text{Image}(F))$.

(H2) Σ is a compact connected Lie group with bi-invariant metric (Definition 2 of v1.0), and $\Phi: \mathbb{R}^n \rightarrow \Sigma$ is the metric projection, idempotent on Σ (Definition 3 of v1.0).

Theorem 1' (Amended Algebraic Collapse)

Let $\mathcal{A}_0 = \Phi(\text{Image}(F)) \subseteq \Sigma$ be the set of constrained outputs, and let

$$\mathcal{A} = \langle \mathcal{A}_0 \rangle = \{a_1 \oplus \cdots \oplus a_m : m \geq 1, a_i \in \mathcal{A}_0\}$$

be the sub-semigroup of Σ generated by the constrained outputs. Then:

(i) [Algebraic closure, corrected] $\mathcal{A}$ is a sub-semigroup of G by construction. Its closure $\text{cl}(\mathcal{A})$ is a compact sub-semigroup of Σ , hence a subgroup of Σ (a classical result for compact groups), and therefore a compact Lie subgroup of Σ with Lie algebra $\mathfrak{h} \subseteq \mathfrak{g}$, by Cartan's closed-subgroup theorem.

(ii) **[Countable reachable algebra]** Under (H1), $\mathcal{A}$ is at most countable: it is the union over $m \geq 1$ of the m -fold product sets $\mathcal{A}_0^m$, each countable (finite products over a countable set), and a countable union of countable sets is countable. Consequently, whenever $\dim \Sigma \geq 1$, $\mathcal{A}$ has Haar measure zero in Σ . The collapse is therefore *from the uncountable continuum Σ to a countable reachable algebra whose closure is a compact Lie subgroup of dimension $d'' \leq d$* — a countably-infinite structure in general, not a finite one.

(iii) **[Subgroup caging — the corrected collapse statement]** Suppose the constraint layer is designed so that $\mathcal{A}_0 \subseteq H$ for some closed proper subgroup $H \subsetneq \Sigma$ with $\dim H = d'' < d$. Then $\mathcal{A} \subseteq \text{cl}(\mathcal{A}) \subseteq H$. That is: *every constrained output, and every finite algebraic combination of constrained outputs, remains confined to the designed lower-dimensional subgroup H — regardless of the model's nondeterminism, sampling temperature, or run length*. The deterministic layer cages the reachable algebra of the generative system inside a designed subgroup.

(iv) **[Enumerability and certificates]** $\mathcal{A}$ is at most countable and therefore formally enumerable. Membership in $\text{cl}(\mathcal{A})$ is verifiable numerically to any precision $\delta > 0$ (the δ -effective framework of Corollary 3), which is the practical verification certificate.

Proof. (i) $\mathcal{A}$ is closed under $\oplus$ by definition of generated semigroup. As a subset of the compact group Σ , $\text{cl}(\mathcal{A})$ is compact; it is a sub-semigroup by continuity of μ . A closed sub-semigroup of a compact group is a subgroup (for each $a \in \text{cl}(\mathcal{A})$, the sequence $a, a^2, a^3, \dots$ has a convergent subsequence whose limit is the identity, which yields inverses). Cartan's theorem then makes $\text{cl}(\mathcal{A})$ a Lie subgroup. (ii) Counting argument as stated; Haar measure zero follows since a countable subset of a manifold of positive dimension is null. (iii) H is a closed sub-semigroup of Σ containing every generator $a \in \mathcal{A}_0$; hence $\mathcal{A} = \langle \mathcal{A}_0 \rangle \subseteq H$, and $\text{cl}(\mathcal{A}) \subseteq H$ because H is closed. (iv) follows from (ii) and compactness of $\text{cl}(\mathcal{A})$. ■

Remark. The caging property (iii) is the honest formal counterpart of the paper's architectural thesis: it is precisely the mathematical statement of a deterministic layer bounding a probabilistic generator's reachable behavior — the bifurcated-governance design. It is also a *stronger* safety statement than the retracted finite-generation claim, because it bounds all algebraic combinations, not merely enumerated outputs.

Entropy component (retained from v1.0). The data-processing inequality gives $H(\Phi(F)) \leq H(F)$ with equality iff Φ is injective almost everywhere on the support of F (in particular when the outputs already lie in Σ). This part of v1.0 Theorem 1(iii) is correct and stands.

B. Defects Found and Disposition

D1 — Theorem 1(i), algebraic closure (v1.0 proof, §5.4). The proof needed to show $\mu(a, b) \in \mathcal{A}$ for $a, b \in \mathcal{A} = \Phi(\text{Image}(F))$. The v1.0 argument — " Φ is idempotent on Σ , so $\Phi(\mu(a, b)) = \mu(a, b)$; closure follows from the group axioms" — establishes only that $\mu(a, b)$ lies in $\Sigma = \text{Image}(\Phi)$, not that it lies in $\Phi(\text{Image}(F))$: a product of two projected outputs need not itself be any single projected output. Closure of the raw image is false in general. **Fix:**

the object is redefined as the generated semigroup $\mathcal{A} = \langle \Phi(\text{Image}(F)) \rangle$, making (i) true by construction (Theorem 1' (i)).

D2 — Theorem 1(ii), finite generation (v1.0 statement and proof). The exponential-map surjectivity theorem for compact Lie groups states that every element of Σ equals $\exp(Y)$ for *some* $Y \in \mathfrak{g}$. It does **not** state that every element is a finite product of the fixed finite set $\{\exp(X_1), \dots, \exp(X_k)\}$: $\exp(Y)$ for $Y = t_1 X_1 + t_2 X_2$ is not $\exp(t_1 X_1) \cdot \exp(t_2 X_2)$ in general (the discrepancy is the Baker–Campbell–Hausdorff series). Finite products of a fixed finite set form a countable set, which cannot equal Σ when $\dim \Sigma \geq 1$. The v1.0 statement is internally inconsistent with its own part (iv). **Fix:** finite generation is retracted; the true, provable structure is the countable generated semigroup with closure a compact Lie subgroup (Theorem 1'(ii)), with the collapse quantified as countability plus subgroup confinement, not finiteness.

D3 — Corollary 2, entropy bound. The bound $H(F) - H(\Phi(F)) \geq (n - d) \cdot \log(1/L)$ was asserted without derivation, and is inconsistent with Theorem 1(iii)'s equality case: when F is supported on Σ (Φ acts as the identity there), the entropy reduction is zero while $(n - d) \cdot \log(1/L) > 0$ whenever $L < 1$. **Fix:** retracted as stated; retained as an open conjecture under additional support assumptions. The provable replacement is the metric-entropy bound of Corollary 2' below.

D4 — Corollary 1, decidability claim. v1.0 claimed output verification is "a decidable problem in finitely-generated group theory (Dehn, 1911)." The word problem for finitely presented groups is undecidable in general (Novikov; Boone) — the citation does not support the claim. **Fix:** retracted; replaced by the honest certificate of Theorem 1'(iv): countable enumeration plus numerical membership verification in $\text{cl}(\mathcal{A})$ to precision δ .

D5 — Figure 3 caption. "Trajectories converge to a finite set of nodes" overstates the corrected result. **Fix:** read "a countable set of nodes whose closure is a compact Lie subgroup of dimension $d'' \leq d$."

D6 — §8.2, hallucination containment. Containment is presented as an established property. It is a *conditional* property: it follows only if the composed projection-and-validation operator satisfies the contraction hypothesis of Theorem 2, which is a property of a deployed pipeline to be established by measurement. **Fix:** all §8.2 containment claims are to be read as conditional on the verified contraction hypothesis, to be established empirically by the scheduled validation trials. Evidence first, status second.

C. Corrected Corollary

Corollary 2' (Metric-entropy bound, proved)

Let $K \subseteq \mathbb{R}^n$ be compact and contain the support of $F(\cdot, t)$ at a fixed t , and suppose Φ is L -Lipschitz on K . Then for every $\varepsilon > 0$,

$$\log N(\Phi(K), \varepsilon) \leq \log N(K, \varepsilon/L),$$

where $N(S, \epsilon)$ denotes the minimum number of ϵ -balls covering S (the metric-entropy / covering-number function). The constraint cannot increase metric entropy at resolution ϵ beyond the entropy of the unconstrained output at the finer scale ϵ/L .

Proof. Let $\{B(x_i, \epsilon/L)\}$ be a minimal (ϵ/L) -cover of K . By L -Lipschitz continuity, $\{B(\Phi(x_i), \epsilon)\}$ covers $\Phi(K)$. Hence $N(\Phi(K), \epsilon) \leq N(K, \epsilon/L)$; take logarithms. ■

D. What Stands Unchanged

The following are unaffected by this amendment and stand as written in v1.0: the background and related-work survey (§2); the manifold construction, constraint-surface definition, projection operator, and curvature computations (§3), including the correctly stated bi-invariant curvature formula $K(X,Y) = \frac{1}{4}\|[X,Y]\|^2 \geq 0$; the stochastic-field formalization and entropy identities (§4); the Lyapunov stability conditions, gradient-flow characterization, and Banach fixed-point statement (§6), all of which are correctly stated as conditional results; the Hybrid Hemisphere Architecture (§7); and the discussion (§9–10), read subject to D6.

E. Status

With these corrections, the manuscript's central result is the caging theorem: a deterministic manifold layer confines the reachable algebra of a nondeterministic generator to a countable, lower-bounded-dimensional structure inside a designed subgroup. The validation of the applied claims (containment, effective determinism) is pending the scheduled measurement trials, per the receipt-first doctrine. This amendment is disclosed in full per the open-book method.

This software is a prototype and is provided for educational and research purposes only. It is not intended for production use, commercial deployment, or safety-critical environments. All systems are experimental and may contain defects on them.

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